PROGESTAGEN- ONLY ORAL CONTRACEPTIVES:

Comparison of the Metabolic Effects of Levonorgestrel and Norethisterone

Ball, MJ*, Ashwell, E, Gillmer, MDG

Department of Obstetrics and Gynaecology, John Radcliffe Hospital, Oxford
* Department of Human Nutrition, University of Otago, Dunedin, New Zealand

ABSTRACT

A 6-month single-blind study compared the use of a progestagen-only oral contraceptive containing norethisterone 350μg/day (NE 350) with one containing levonorgestrel 30μg (LN 30), to assess the metabolic effects.

At the end of 6 months, there were no significant differences between the two groups with respect to plasma cholesterol, lipoproteins including HDL subfractions, triglycerides or glucose concentration. Levels of fibrinogen, plasminogen, Factor VII, Factor X and antithrombin III were also similar. Women changing from a combined oral contraceptive to LN 30 showed a significant fall in Factor X.

Mean blood pressure fell on LN 30 by 7/9 mmHg, but the 6-month reading did not differ significantly from that in women on NE 350.

Acceptability, and the metabolic effects of the two preparations were similar in this study. Further larger studies are warranted.

INTRODUCTION

Widespread, prolonged use of oral contraceptives (OCs) has given rise to concern about their influence on the risk factors for cerebro- and cardio-vascular disease. Use of combined OCs, both present and past (1, 2, 3), is associated with a small increase in risk for cardiovascular disease. This may be mediated by the effects of the preparations on lipid metabolism, glucose tolerance and the coagulation system.

Initial drug safety reports indicated the risk was related to the oestrogen dose (3, 4). However, the progestagen component also has an effect (5), depending on the type and dose. The RCGP study published in 1977 (6) suggested a dose-response relationship between norethisterone (NE) and hypertension. Meade et al. (4) reported a significant positive association between the NE content and deaths from stroke and coronary heart disease (CHD) in women using combined OCs, but not of hypertension, and a possible excess of deaths from stroke with combined OCs of high (250μg) levonorgestrel (LN) content.

Most studies have looked at the metabolic effects of combined OCs. With these preparations the alteration in coagulation and lipid metabolism depends on the interaction of the two components. For example, oestrogens tend to increase plasma triglyceride, VLDL and high density lipoprotein cholesterol (HDL–C) (7, 8), whereas progestagens can have the opposing effects depending on their relative biological activity and anti-oestrogenic/ androgenic properties. Androgenic and progestational steroids tend to decrease HDL–C (9). An increased CHD risk is associated with a high plasma concentration of total cholesterol and LDL–C, reduced HDL–C (10, 11) and, in some circumstances, high triglycerides (TG) (12). OCs may also adversely affect coagulation and reduce glucose tolerance (7).

The use of progestagen-only pills (POP) is increasing, but there is little reliable information about their metabolic effects and their influence on cardiovascular disease risk factors.

Submitted for publication February 20, 1991
Accepted for publication July 11, 1991

This study investigates the effects of two POPs containing NE and LN, respectively, on lipoprotein levels, glucose tolerance, coagulation factors and blood pressure.

METHOD

Women aged 17-41 who requested oral contraception were recruited into a single-blind parallel study to compare patient acceptability and metabolic effects of norethisterone 350 μg/day (NE 350) and levonorgestrel 30 μg/day (LN 30). The women were randomised to treatment and stratified according to prior OC use. NEW patients were defined as those who had not used an oral contraceptive or hormone therapy for at least 3 months. SWITCH patients were those changing from a low-dose combined OC, usually because of possible side effects or older age.

Women were not recruited if they were hypertensive (diastolic BP >100mmHg, systolic BP > 140mmHg), smoked more than 20 cigarettes per day, or were diabetic. The duration of treatment was six continuous 28-day periods. Assessment was performed at baseline, and month 6 when weight and blood pressure were recorded and a fasting venous blood sample was taken. Diary cards were also completed of episodes of bleeding and menstrual cramps.

METHODS OF ANALYSIS

The total cholesterol concentration and the cholesterol concentration in the lipoprotein fractions was measured enzymatically (13). HDL–C was measured after precipitation of LDL and VLDL with heparin and manganese (14). HDL subfractions in the supernatant were measured by precipitating HDL_2 with dextran sulphate and measuring the HDL_3 subfraction remaining in solution (15). The quantity of cholesterol in the various subfractions was calculated by subtraction. Plasma triglyceride concentration was measured enzymatically using a kit obtained from Boehringer. Plasma glucose was measured enzymatically using glucose oxidase (16). Glycosylated Hb was measured using thiobarbituric acid (17).

Venous blood samples for analysis of coagulation factors were taken,mixed with trisodium citrate 0.109M (9:1 v/v), and centrifuged immediately at 2,000xg for 15 min to obtain platelet-poor plasma which was snap-frozen and stored as aliquots in plastic tubes with liquid nitrogen.

Factor VII clotting activity was measured by a one-stage assay based on the prothrombin time using an artificially depleted substrate plasma (18).

Factor X clotting activity was measured using Russell's viper venom and a factor X and VII depleted charcoal-absorbed bovine plasma (18).

Fibrinogen concentration was measured using the Clauss assay (19).

Antithrombin III (AT-III) activity was measured by an amidolytic microtitre assay based on the method of Odegaard (20) using a specific chromogenic substrate, S-2238 (Kabi Diagnostica, Stockholm),and the British Standard for coagulation factors from the National Institute for Biological Standards and Controls.

Plasminogen was measured chemically (21)

Switch patients were analysed separately from new patients for laboratory data because of expected effects of the previously used combined OCs. Six-month data for all women were then combined and a comparison made between the results for women who received LN and those who had received NE.

All continuous variables were tested for normality prior to analysis using BMDP 2D. Changes from baseline within group were analysed using the Students paired t test or Wilcoxon signed rank test. Comparisons between treatments were made using

the two-sample t-test for normally distributed variables or the Mann-Whitney U test for the non-parametric.

RESULTS

Fifty-one women were recruited: five did not return for follow-up and their data was excluded. The mean age of both groups was 29 years (range 17-41 years). The two groups were well matched for height, weight and body mass index (BMI). The mean (SD) BMI was 22.4 (2.3) in those randomised to receive NE 350 and 21.7 (3.4) in those who received LN 30. Ten of the 46 patients were smokers, who continued to smoke during the study. Of those who completed the study, 2 new OC users in each group were smokers, as were 2 switch subjects on LN and 1 on NE. All but one woman consumed some alcohol.
Fifteen women in each group were nulliparous.

Twenty-three women were new OC users. Twenty-three were switched from combined OCs, which they had taken for one to 192 months, onto the trial preparation; 12 receiving NE 350 and 11 LN 30.
Seven new subjects reported previous oligo- or amenorrhoea.
Nine women withdrew after three months (1 on NE 350 , 8 on LN 30); 3 no longer needed contraception; 1 had menstrual cycle problems, 1 had a coil inserted, and 4 were lost to subsequent follow-up. Four women in each group came for weight and blood pressure measurement at 6 months but did not return for a fasting blood sample.

Weight, Blood Pressure and Acceptability
Women in the NE 350 group tended to be slightly heavier than those in the LN 30 group. Weight increase was slight but at the end of six months the mean weights were 59.8 kg and 55.3 kg, respectively (significantly different, p = 0.04) - see Table I. Systolic BP decreased from baseline on both treatments, but only marginally so on NE 350. On LN 30 the mean systolic BP had fallen by 7 mmHg (p=0.027) after 6 months treatment, and the diastolic phase IV and V by 9 mmHg (p<0.01).

Table I
Changes in weight (kg) and blood pressure (mmHg) from baseline of women on progestagen-only pills containing norethisterone (NE 350) or levonorgestrel (LN 30); Mean (SD)

		NE 350 (n=23)		LN 30 (n=23 at 3 mths 16 at 6 mths)		Difference between groups (p)
Weight	(mth 6)	0.0	(1.7)	0.6	(1.9)	0.30
Systolic BP	(mth 6)	-2.8	(8.8)	-7.1	(11.6)	0.20
Diastolic BP (phase IV)	(mth 6)	-3.2	(8.0)	-9.0	(9.8) *	0.06
Diastolic BP (phase V)	(mth 6)	-2.4	(8.5)	-8.8	(8.7) *	0.03

* change in Korotkoff phase IV and V from baseline statistically significant, p<0.01, paired t test.

This decrease in diastolic V was signficantly larger (p=0.03) than for the NE 350 group (2.4 mmHg fall), and at 6 months the mean diastolic pressure was 65.8 mmHg on LN 30 vs 73 mmHg on NE 350 (p = 0.057).

The mean number of episodes of bleeding per calendar month was 1.28 on LN 30 (range 0 - 2.3) and 1.24 on NE 350 (range 0.4 - 2). Menstrual cramps decreased slightly on treatment. Cycle control at 3 and 6 months was similar; two women on NE 350 and three on LN 30 reported 'poor' cycle control. The total number of reports of possible adverse events was similar.

Laboratory Data. Pre-treatment results were not significantly different for the two groups. Separate analyses for new and switch patients are shown in Tables II - IV.

Coagulation Factors

Mean factor VIIc levels fell significantly by almost 20% (p=0.004) in women switched to NE 350, but the fall was not significant in those switched to LN 30. In new POP users,the decrease on NE 350 was 0.11 IU/ml (p=0.15) compared to no change on LN 30, and at month 6,levels were significantly lower on NE 350 (p=0.013). In switch subjects, Factor X fell significantly from baseline on both treatments, but did not change significantly in the new users. There was a significant decrease in antithrombin III (p = 0.04) in the switch women on NE 350, but no change on LN30. Fibrinogen and plasminogen levels did not change significantly.

Table II. Levels of coagulation factors in women switching from combined pills and in new users; Mean (SD) - IU/ml

NE 350

	Switch from OC (n=10)		New User (n=10)	
	Pre-	6 mth	Pre-	6 mth
Factor VIIc	1.16 (0.21)	0.93 (0.19)††	0.95 (0.19)	0.84 (0.13) *
Factor X	1.33 (0.39)	0.99 (0.12)††	0.97 (0.22)	0.99 (0.10)
A-T III	1.08 (0.05)	0.96 (0.05)†	1.16 (0.03)	1.08 (0.05)
Fibrinogen (g/L)	1.85 (0.15)	1.77 (0.18)	1.85 (0.17)	1.80 (0.17)
Plasminogen	1.30 (0.13)	1.29 (0.08)	1.18 (0.08)	1.21 (0.09)

LN 30

	Switch from OC (n=7)		New User (n=6)	
	Pre-	6 mth	Pre-	6 mth
Factor VIIc	1.02 (0.21)	0.89 (0.15)	1.04 (0.13)	1.04 (0.06) *
Factor X	1.12 (0.23)	0.85 (0.25)†	0.96 (0.13)	0.95 (0.12)
A-T III	1.09 (0.06)	1.10 (0.09)	1.08 (0.04)	1.07 (0.06)
Fibrinogen (g/L)	1.78 (0.21)	1.75 (0.20)	2.03 (0.37)	1.81 (0.16)
Plasminogen	1.34 (0.11)	1.32 (0.18)	1.05 (0.12)	0.98 (0.05)

Change from baseline significant † p<0.05 †† p<0.02, Wilcoxon signed rank test.
* difference between NE and LN statistically significant p = < 0.05

Reference values -		
	Factor VIIc	0.5 - 2 IU/ml
	Factor X	0.5 - 2 IU/ml
	Antithrombin III	0.8 - 1.2 IU/ml
	Fibrinogen	1.5 - 4.0 g/L
	Plasminogen	0.5 - 1.5 IU/ml

All parameters appeared similar after 6 months use of the two OCs (see Table V)

Lipids and lipoproteins

The mean plasma cholesterol decreased from baseline in both new and switch groups on both treatments, but this was not statistically significant, and there was no difference between treatments. All but 2 women had a cholesterol level below 5.5 mmol/l at the end of the study: one woman with a level exceeding this was a new user on NE whose initial plama cholesterol was 5.5 and this increased by 1 mmol/L due to a rise in HDL-C. LDL–C did not change significantly in any group. LDL-C exceeded 3.5 mmol/L in only one woman, in whom it was 4.4 mmol/l pre-treatment and 4.0 mmol/L after 6 months on NE.

New patients on both pills showed a small mean increase of about 0.1 mmol/L in HDL–C. Switch patients on NE 350 also showed a mean fall in HDL–C of 0.11 mmol/L which was significantly different (p=0.03) from the increase of 0.14 mmol/L on LN 30. In new users,equal numbers of women showed an increase and a decrease. New patients on LN 30 showed a greater change in HDL2 than switch patients. The changes ranged from +5% to -30% in women using NE 350 and from +10% to - 28% in women on LN 30, but the trend was an increase in women switching to LN 30. The changes in HDL3 tended to be in the reverse direction to those of HDL2. The HDL2/HDL3 ratio fell in new patients on LN 30, in contrast to the small increase on NE 350, but the difference was not statistically significant. The HDL2/LDL ratio was not significantly altered. The fall in plasma triglycerides in women switching to LN 30 approached significance (p = 0.6), but at month 6 there was no obvious difference between users of the two preparations (Table V).

Table III Plasma lipid and lipoprotein levels; Mean (SD), mmol/L

	NE 350 (n = 21)			
	Switch from combined OC		New Users	
	Pre-	6 mth	Pre-	6 mth
Cholesterol	4.52 (0.82)	4.12 (0.85)	4.33 (0.73)	4.13 (0.98)
HDL–C	1.55 (0.37)	1.44 (0.22)	1.60 (0.26)	1.66 (0.44)
HDL2	0.35 (0.27)	0.29 (0.15)	0.40 (0.19)	0.27 (0.19)
HDL3	1.22 (0.24)	1.15 (0.22)	1.17 (0.19)	1.15 (0.58)
LDL–C	2.60 (0.90)	2.26 (0.73)	2.32 (0.65)	1.98 (0.94)
VLDL–C	0.44 (0.07)	0.45 (0.11)	0.40 (0.23)	0.47 (0.26)
Triglyceride	1.39 (0.42)	1.05 (0.45)	1.20 (0.46)	0.93 (0.27)
	LN 30 (n = 13)			
	Switch from combined OC		New Users	
	Pre-	6 mth	Pre-	6 mth
Cholesterol	3.93 (0.90)	3.82 (0.33)	4.60 (0.68)	4.19 (0.70)
HDL-C	1.49 (0.02)	1.56 (0.29)	1.45 (0.16)	1.60 (0.21)
HDL2	0.24 (0.18)	0.35 (0.22)	0.50 (0.15)	0.41 (0.22)
HDL3	1.29 (0.06)	1.24 (0.12)	0.92 (0.09)	1.14 (0.30)
LDL-C	2.19 (0.75)	1.94 (0.51)	2.36 (0.87)	2.22 (0.57)
VLDL-C	0.32 (0.20)	0.31 (0.12)	0.64 (0.18)	0.47 (0.30)
Triglyceride	1.35 (0.28)	0.93 (0.52)	1.18 (0.65)	0.81 (0.35)

Glucose and Glycosylated Hb

The plasma glucose concentration fell significantly from baseline in new patients on

NE 350 (mean fall of 1 mmol/L; p=0.02) which was significantly different (p=0.03) from the non-significant change on LN 30. Five women had a level exceeding the reference value pre-study but the value was below 8 mmol/l in all cases: two were randomised to LN and 3 to NE. At 6 months, just 2 women on LN had raised levels. Glycosylated haemoglobin was not significantly different; but was slightly raised in 1 woman randomised to each of LN and NE, and after 6 months was raised in 2 women on LN and 3 on NE.

Table IV. Plasma glucose and glycosylated Hb (HbA1c) levels; Mean (SD)

	NE 350			
	Switch from OC (n=10)		New User (n=10)	
	Pre-	6 mth	Pre-	6 mth
Glucose	5.3 (1.0)	4.5 (1.5)	5.5 (0.8)	4.5 (0.5)†
HbA1c (%)	6.8 (0.8)	6.4 (1.1)	6.6 (0.7)	7.1 (1.3)
	LN 30			
	Switch from OC (n=6)		New User (n=6)	
	Pre-	6 mth	Pre-	6 mth
Glucose	5.5 (0.5)	5.4 (1.2)	5.3 (0.9)	5.4 (1.0)
HbA1c (%)	7.1 (0.6)	6.6 (1.2)	6.7 (0.3)	6.7 (0.8)

† change from baseline statistically significant, $p < 0.05$
Reference range - fasting glucose <6.5 mmol/l, glycosylated Hb <7.8%

The combined results of all users (new plus OC switch) after 6 months on both POPs is shown in Table V. There was no significant difference between the results for coagulation factors, lipoproteins or glucose in women on the two preparations.

Table V. Comparison of laboratory results at the end of 6-months POP use; Mean (SD)

	NE 350 (n = 20)	LN 30 (n = 13)	Difference
Factor VIIc (IU/ml)	0.89 (0.17)	0.93 (0.15)	NS
Factor X (IU/ml)	0.99 (0.10)	0.90 (0.20)	NS
Antithrombin III (IU/ml)	1.02 (0.17)	1.09 (0.17)	NS
Fibrinogen (g/L)	1.78 (0.54)	1.72 (0.40)	NS
Plasminogen (IU/ml)	1.25 (0.25)	1.18 (0.37)	NS
Total Cholesterol (mmol/L)	4.12 (0.89)	4.01 (0.74)	NS
LDL (mmol/L)	2.20 (0.69)	2.07 (0.53)	NS
VLDL (mmol/L)	0.43 (0.32)	0.39 (0.22)	NS
HDL (mmol/L)	1.55 (0.35)	1.58 (0.25)	NS
HDL_2/HDL_3	0.32 (0.32)	0.33 (0.20)	NS
LDL/HDL2	10.57 (9.75)	7.61 (5.90)	NS
Triglycerides (mmol/L)	0.99 (0.37)	0.87 (0.42)	NS
Glucose (mmol/L)	4.77 (0.60)	5.39 (1.07)	NS
Glycosylated Hb (%)	6.76 (1.22)	6.67 (0.97)	NS

NS = difference not statistically significant (Mann-Whitney U test)

DISCUSSION

Progestagen-only pills (POPs) are now quite widely used, particularly by older women. However, there is relatively little information on the influence of progesterone and its derivatives on plasma lipoproteins and carbohydrate metabolism in women of reproductive age, and even less on the effects on coagulation factors. This may be due to greater difficulty in recruiting large numbers of women into a POP study than a combined OC study, because less women wish to use this type of pill and the cycle control is less good. Most data have thus been extrapolated from studies of progestagens combined with estrogens in various OC preparations.
The two progestagens examined in the current study are the commonly used 19-nortestosterone-related compounds - levonorgestrel (LN) and norethisterone (NE). LN is a gonane derivative and NE an estrone derivative. Gonanes have up to ten-fold greater progestational potency than estranes by weight - hence the different doses used - and have greater androgenic effect (22). Population studies suggest an association between androgenic activity and increased CHD risk (23). Gonane progestogens tend to induce a lipid profile similar to that of men. This is contrary to the effect of oestrogens which usually decreases LDL and increases HDL; a potentially beneficial effect.

In this study we investigated the effects of progestagens alone under two different circumstances - firstly in women starting to use POPs and secondly in women switching to a POP from a combined OC. In the latter group the changes during the study thus reflect the difference between the effects of a combined OC containing 30-35μg oestrogen (EE) and a progestagen (LN in all but 3 women where it was NE) and the POP. The oestrogen will have tended to increase serum triglycerides and HDL, and to decrease antithrombin III and have attenuated the progestagen effect (21). This allows a comparison of practical value, which has not been studied previously in this longitudinal manner. However, the numbers in each individual group are small and this may result in small effects being missed or exaggerated.

The 6-month results of new and switch patients were also combined in one analysis to allow comparison of the two POPs effects in a larger group. It was felt that the metabolic influence of any previously used combined OC would have disappeared after 6 months, although this may be the minimum time required.

Effect on LDL

The effects of progestagens on LDL-C are thought to be minimal, although high doses of dl-norgestrel (1800μg/day) may increase LDL triglyceride in oophorectomised women (23). Combined OCs may increase LDL cholesterol and in particular LDL triglyceride content, and Wahl (8) found the highest LDL in women on pills with the lowest oestrogen and the strongest anti-oestrogenic progestagen. Ahren (24) found that women on 30μg EE and 150μg LN had a significantly raised LDL triglyceride at 3 and 6 months - returning towards baseline at 12 months. In our study, LDL-C did not increase on either progestagen; and the lack of this effect with the androgenic progestagen LN may result from the low dose of this POP.

Effect on plasma triglyceride

Studies have variously suggested that progestagens have no effect on plasma triglycerides or lower them slightly. Women on combined OCs often show a rise in triglycerides, due to the oestrogen component. The more androgenic gonane progestagens oppose this (7, 25). Progestagens may reduce triglycerides in people with hyperlipidaemia (26) and some studies have indicated a fall with NE and norgestrel in modest doses.

Larsson-Cohn (27) found serum triglycerides were related to oestrogen/LN ratio, and the effect of NE is also likely to be dose-dependent. The mechanism of the effect is unclear, but studies in the LN-treated rat indicate that VLDL clearance is enhanced; and hepatic lipase activity is stimulated by androgenic progestagens (29).

In our switch subjects,plasma triglycerides fell by up to 40% reflecting the removal of the oestrogen. The mean fall was twice as great in women who switched to LN. New patients showed a small fall, again greater in those taking LN. This was not statistically significant, but the trend would concur with that in other studies.

Effect on HDL-C

The lipoprotein change observed most consistently with NE and LN use is a dose-dependent reduction in HDL cholesterol, but in women using the combined OC this is moderated by the effect of oestrogens which increases HDL (21). Gonane progestagens appear to reduce HDL most. Wahl (8) found women on high oestrogen combined OCs had less reduction in HDL-C when the preparation contained NE. In women on lower oestrogen OCs, HDL–C is variably reduced (28). In the cross-sectional study by Godsland et al. (29), HDL-C was highest in women on low-dose monophasic NE OCs and lowest in those on monophasic LN combinations. Reduction in HDL–C was not seen in our study when women started on LN or NE, or switched from a combined OC. HDL subfractions measured as total HDL–C may be misleading. It is HDL_2 that correlates inversely with CHD, and a rise in HDL_3 levels, which tends to occur with combined OCs, may obscure a fall in HDL_2. HDL subfractions have not been measured in many studies, but gonane progestagens (e.g. LN) seem to reduce HDL_2 more than estrane progestagens. This effect may be mediated by changes in the activity of the enzyme lipoprotein lipase which hydrolyses HDL_2 phospholipids (30). Estrane progestagens in combined OC may cause a reciprocal increase in HDL_3, which is not seen with high-dose LN-containing preparations (29,31). In our study,mean levels of HDL_2 fell in both groups of new users. The fall in mean HDL_2 in women switching from combined OC to NE 350 would seem to reflect the loss of oestrogen intake, as oestrogens raise HDL_2.

Mean HDL_3 rose slightly in our new patients on LN but the ratio of HDL_2 to HDL_3 was not significantly altered. After 6 months there were no significant differences between the two groups with respect to HDL, HDL_2 or HDL_2/HDL_3 ratio. This does, however, need to be examined in further studies as the small numbers may have meant that a small difference could have been missed. Wynn (31) found a slight reduction in HDL_2 in women taking POPs containing 250μg NE and 70μg dl norgestrel, with a slightly lower dose of active LN than in our study.

Effect on carbohydrate risk factors

Mild impairment of glucose tolerance has been associated with increased CHD risk (32), and increased plasma glucose levels in response to a glucose load are found in some women taking combined OCs.

Deterioration in glucose tolerance appears greatest with a high oestrogen dose (>70μg) (7), which reduces the insulin response for a given plasma glucose. However, progestagens can diminish glucose tolerance, although oestrogen may potentiate this effect. Women on combined OCs with LN and also on LN alone tend to have reduced glucose tolerance and increased insulin secretion. A large cross-sectional study of women receiving 9 different OCs (29) revealed that the response of plasma glucose and insulin levels to an oral glucose tolerance test was greater in OC users than non-users. The effect was greater with OC combinations containing LN than for those containing NE or desogestrel. Women on LN POPs also had an

increased response. Spellacy, (33) reported a significantly higher blood glucose, plasma insulin and weight gain after 18 months on 75 μg norgestrel. A progressive decline in glucose tolerance over time has been shown with low-dose LN OCs (34) but was not seen with a low-dose NE preparations (35). Oakley and Rostron (36) also found a difference between combined OCs with EE 30-35 and LN 150 or NE 500/1000, with a decrease in glycosylated Hb on the latter.

Fasting plasma glucose is an insensitive measure of small changes in glucose tolerance, but as mean levels fell in all groups, significantly in new users of NE 350, and the fact that glycosylated Hb, which is an index of mean blood glucose, did not rise, it would seem unlikely that glucose tolerance had deteriorated significantly. A better assessment of glucose tolerance would have been an OGTT with glucose and insulin measurements, but the women in this study were not keen to undergo this procedure. It is hoped that this can be performed in future studies.

Effect on coagulation factors

Raised levels of factor VIIc and fibrinogen appear to be related to increased CHD risk (37). Combined OCs have complex effects on coagulation factors with oestrogen having a dose-response effect in increasing Factor VIIc and fibrinogen levels (38). These effects tend to be pro-coagulant, but may be matched by changes in the fibrinolytic system (38,39,40), particularly anti-thrombin III activity.
In our study, fibrinogen levels were not increased by the POPs. Factor X levels fell in the switch patients, reflecting the discontinuation of the oestrogen dose, and Factor VIIc fell significantly in women who switched to NE, but not those who used LN. This may reflect a slightly different effect of the two progestagens. Factor VIIc was also lower in new NE users. Antithrombin III, an important inhibitor of coagulation, which is increased by oestrogens, fell slightly in women switching from combined OC to NE but not LN. This parallels the effect of the pro-coagulant factor VIIc. At 6 months, however, the coagulation activities were similar on the two POPs and it would appear that neither has effects which markedly alter the clotting cascade.

Effect on blood pressure

A link between progestagen content and hypertension has been demonstrated in some epidemiological studies (6) . The finding was not, however, confirmed by Meade et al. (4). POPs do not appear to cause hypertension but the data is not extensive. A 10-year open, multicentre study on NE 350 revealed no consistent changes in blood pressure or weight (41). Another study by Wilson et al. (42) also found no change in blood pressure in women receiving NE 350 or 75 μg norgestrel. Mean blood pressure fell in our study, but this is often seen with repeated longitudinal measurements. Data relating this to any OC change is inconclusive.

The group of women studied are not a high risk group for CHD. However, POPs are more likely to be prescribed for older women, and it is known that increases in cholesterol and LDL-C may have some adverse effect even if the baseline levels are quite low (43), and that mild impairment of glucose tolerance is undesirable. From this study, use of the POPs containing 30 μg LN and 350 μg NE would seem to be associated with little lipid or carbohydrate mediated CHD risk, and to have less effect on coagulation than other preparations and this is reassuring. Although it is difficult to recruit women to studies of POPs, larger studies will be needed to confirm the findings and to reveal any small differences between the preparations which may not have been obvious in this study.

Acknowledgements. Dr S Machin, Middlesex Hospital, kindly performed the coagulation assays. The lipoprotein assays were performed in Dr J Mann's laboratory.

REFERENCES

1. Vessey M, Villard-Mackintosh L, McPherson K, Yeates D. Mortality among oral contraceptive users; 20 year follow up. Brit Med J 1989; 299: 1487-91
2. Slone D, Sharpiro S, Kaufman D et al. Risk of myocardial infarction in relation to current and discontinued use of oral contraceptives. N Engl J Med 1981; 305: 420-24
3. Mann J I, Inman W H. Oral contraceptives and death from myocardial infarction. Brit Med J 1975; 2: 245
4. Meade W, Greenberg G, Thompson S. Progestagens and cardiovascular reactions associated with oral contraceptives and a comparison of the safety of 30μg and 50μg oestrogen preparations. Brit Med J 1980; 280: 1157-61
5. Kay C R. The Royal College of General Practitioners' oral contraception study. Obstet Gynecol 1984; II: 759-86.
6. Royal College of General Practitioners. Effect on hypertension and benign breast disease of progestagen component in combined oral contraceptives. Lancet 1977; 1: 624.
7. Wynn V et al. Comparison of effects of different combined oral contraceptive formulations on carbohydrate and lipid metabolism. Lancet 1979; 1045-49.
8. Wahl T, Walden C, Clock R, et al. Effect of estrogen/progestin potency on lipid/lipoprotein cholesterol. N Engl J Med 1983; 308: 15, 862-67.
9. Silferstolpe G, Gustafson A, Samsioe G. Lipid metabolic studies in oophorectomized women. Effects of three different progestagens. Acta Obstet Gynecol Scand 1979; suppl; 88: 89-93.
10. Dawber T R. The Framingham Study: the epidemiology of atherosclerosis disease. Cambridge, MA: Harvard University Press,1980.
11. Kannel W B. High density lipoproteins. Epidemiologic profile and risks of coronary heart disease. Am J Cardiol 1983; 52: 9B - 12B.
12. Castelli W P. The triglyceride issue. A view from Framingham. Am Heart J 1986; 112: 432-37.
13. Seidel J et al. Improved reagent for the enzymatic determination of serum cholesterol. J Clin Chem 1981; 19: 838-39.
14. Burnstein M, Scholnick H, Morfin R. Rapid method for the isolation of lipoproteins from human serum by precipitation with polyanions. J Lipid Research 1970; II: 583-589.
15. Ononogbu I C, Lewis B. Lipoprotein fractionation by a precipitation method. Clin Chim Acta 1976; 71: 397-402.
16. Trinder P. Determination of glucose in blood using glucose oxidase. Ann Clin Biochem 1969; 6: 24-27.
17. Ross I S, Gibson P F. A semi-automated method for the determination of glycosylated haemoglobin. Clin Chem Acta 1979; 98: 53-59.
18. Giddings JC. Hereditary coagulation disorders: Laboratory techniques. In: Blood Coagulation and Haemostatis - A Practical Guide. JM Thomson, ed. Churchill Livingstone,1987.
19. Von Clauss A. Gerinhungsphysiologische schnellmethode zur bestimmung des fibrinogens. Acta Haematologia 1957; 17: 237.
20. Odegaard ER et al. Heparin co-factor actually measured with an amidolytic method. Thromb Res 1975; 6: 287.
21. Machin S, Mackie I. Coagulation factors. In: Practical Haematology. Charin I, ed. Churchill Livingstone, 1989

22. Dickey R, Stone S. Progestational potency of oral contraceptives. Obstet Gynecol 1976; 47: 106-112.
23. Godsland IF, Wynn V, Crook D et al. Sex, gonadal steroids and plasma lipoproteins. Am Heart J 1987; 114: 1467-70.
24. Ahren T, Victor A, Lithell H et al. Comparison of the metabolic effects of two hormonal contraceptives. Contraception 1981; 24: 451-57.
25. Lipson A, Stoy D, LaRosa JC et al. Progestins and oral contraceptive-induced lipoprotein changes: a prospective study. Contraception 1986; 34: 121-34.
26. Glueck CH, Levy RI, Fredrickson DS. Norethindrone acetate, post heparin lipolytic activity and plasma triglycerides in familial type I,II IV and V hyperlipoproteinaemia. Ann Int Med 1971; 75 : 345- 52.
27. Larsson-Cohn U, Fahraeus L, Wallentin L, et al. Lipoprotein changes may be minimised by proper composition of a combined oral contraceptive. Fertil Steril 1981; 35: 172-79.
28. Van der Vange N, Kloosterboer HJ, Haspels AA. Effects of seven low- dose combined oral contraceptives on high density lipoprotein subfractions. Br J Obstet Gynaecol 1987; 94: 559-67.
29. Gosland I, Crook D, Simpson R et al. The effects of different formulations of oral contraceptive agents on lipid and carbohydrate metabolism. NEJM 1990:323;1375-81
30. Tikkanen M, Nikkila E, Kuusi T, et al. High density lipoprotein 2 and hepatic lipase. J Clin Endocrinol Metab 1982; 54: 113-17.
31. Wynn V, Niththyananthan R. The effects of progestins in combined oral contraceptives with special reference to high-density lipoproteins. Am J Obstet Gynecol 1982; 142: 766-72.
32. Pyorala K, Savolainen E, Lehtovirta E, Punsar S, Siltanen P. Glucose tolerance and coronary heart disease. Helsinki Policemen Study. J Chron Dis 1979; 32: 729.
33. Spellacy W, Buhi W, Birk S. Prospective studies of carbohydrate metabolism in "normal" women using norgestrel for 18 months. Fertil Steril 1981; 35: 167-71.
34. Wynn V. Effect of duration of low dose oral contraceptive administration on carbohydrate metabolism. Am J Obstet Gynaecol 1982;142: 739 -45.
35. Wynn V, Godsland I. Effects of oral contraceptives on carbohydrate metabolism. J Reprod Med 1986; 31: 892-97.
36. Oakley N, Rostron W. Effect of combined oral contraceptives on glycosylated haemoglobin. J Roy Soc Med 1982; 75: 234-36.
37. Meade TW et al. Haemostatic function and ischaemic heart disease: principle results of the Northwick Park heart study. Lancet 1986; 533-37.
38. Meade TW. Oral contraceptives, clotting factors and thrombosis. Am J Obstet Gynecol 1982; 142: 758-61.
39. Sabra A, Bonnar J. Haemostatic system changes induced by 50μg and 30μg estrogen/progestagen oral contraceptives. J. Reprod Med 1983; 28: 85-91.
40. Mammen EF. Oral contraceptives and blood coagulation: A critical review. Am J Obstet Gynaecol 1982; 142: 758-61.
41. Lawson JP. Experience with norethisterone 0.35mg daily as an oral contraceptive. Brit J Fam Planning 1982; 8: 84-89.
42. Wilson E, Crickshank J, McMaster M, Weir R. A prospective study of the effect on blood pressure of contraceptive preparations containing different types and doses of progestogen. Br J Obstet Gynaecol 1984;91:1254-60.
43. Multiple risk factor intervention trial research group: Risk factor changes and mortality results. JAMA 1982; 248 : 1465-77.